MASTER BLASTER QUESTION BANK

FOR JEE I CUET I BOARD & COMPETITIVE EXAMS

CIRCLE

PART-I SOLVED QUESTION ANSWER SERIES

SOLVED QUESTION ANSWER SERIES

Ex.1 Prove that the radii of the circles $x^2 + y^2 = 1$, $x^2 + y^2 - 2x - 6y = 6$ and $x^2 + y^2 - 4x - 12y = 9$ are in A. P.

Sol. Given circles are $x^2 + y^2 = 1^2$ ...(1)

$x^2 + y^2 - 2x - 6y = 6$ ...(2)

and $x^2 + y^2 - 4x - 12y = 9$ ...(3)

Let r_1 , r_2 and r_3 be the radii of the circles (1), (2) and (3) respectively,

then, $r_1 = 1$, $r_2 = \sqrt{(-1)^2 + (-3)^2 + 6} = 4$ and

$r_3 = \sqrt{(-2)^2 + (-6)^2 + 9} = 7$

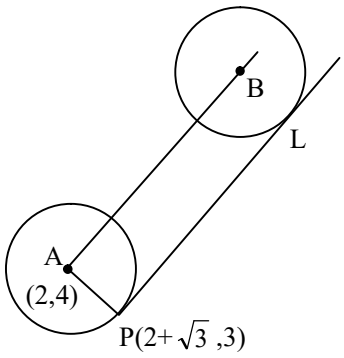
Clearly $r_2 - r_1 = r_3 - r_2$. Hence r_1, r_2, r_3 are in A.P. **Ans.**

Ex.2 The circle $x^2 + y^2 - 4x - 8y + 16 = 0$ rolls up the tangent to it at $(2 + \sqrt{3}, 3)$ by 2 units, assuming the x-axis as horizontal, find the equation of the circle in the new position.

Sol. Given circle is $x^2 + y^2 - 4x + 16 = 0$ ...(1)

Let $P \equiv (2 + \sqrt{3}, 3)$,

Equation of tangent to the circle (1) at $P(2 + \sqrt{3}, 3)$ will be



$(2 + \sqrt{3})x + 3y - 2(x + 2 + \sqrt{3})$

$- 4(y + 3) + 16 = 0$

or $\sqrt{3}x - y - 2\sqrt{3} = 0$ ...(2)

If line (2) makes an angle θ with the positive direction of x-axis,

then $\tan\theta = \sqrt{3}$ $\therefore$ $\theta = 60^\circ$.

Let A and B be the centres of the circles in old and new positions respectively, then

$A \equiv (2, 4)$ and $B \equiv (2 + 2\cos 60^\circ, 4 + 2\sin 60^\circ)$

[$\because$ AB = 2]

thus $B \equiv (3, 4 + \sqrt{3})$

Radius of the circle = $\sqrt{2^2 + 4^2 - 16} = 2$

$\therefore$ Equation of the required circle is

$(x - 3)^2 + (y - 4 - \sqrt{2^2 + 4^2 - 16})^2$

or $x^2 + y^2 - 6x - 2(4 + \sqrt{2^2 + 4^2 - 16})y + 24$

$+ 8\sqrt{2^2 + 4^2 - 16} = 0$ **Ans.**

Ex.3 Obtain the locus of the point of intersection of the tangents to the circle $x^2 + y^2 = a^2$ which include an angle α.

Sol. Let (x_1, y_1) be the point of intersection of a pair of tangents to the circle $x^2 + y^2 = a^2$. The combined equation of this pair of tangents is $SS_1 = T^2$

i.e. $(x^2 + y^2 - a^2)(x_1^2 + y_1^2 - a^2) = (xx_1 + yy_1 - a^2)^2$

or $(y_1^2 - a^2)x^2 - 2x_1y_1xy + (x_1^2 - a^2)y^2 + 2a^2x_1x$

$+ 2a^2y_1y - a^2(x_1^2 + y_1^2) = 0$

If this pair of straight lines includes an angle α, we have

$$\tan\alpha = \frac{2\sqrt{x_1^2y_1^2 - (y_1^2 - a^2)(x_1^2 - a^2)}}{y_1^2 - a^2 + x_1^2 - a^2}$$

Hence more $(x_1^2 + y_1^2 - 2a^2)^2 \tan^2\alpha$

$= 4a^2(x_1^2 + y_1^2 - a^2)$

The locus of (x_1, y_1) is then $(x^2 + y^2 - 2a^2)^2 \tan^2\alpha = 4a^2(x^2 + y^2 - a^2)$ **Ans.**

Ex.4 Prove that the circles $x^2 + y^2 - 4x - 6y + 9 = 0$ and $x^2 + y^2 + 2x + 2y - 7 = 0$ touch. Find the coordinate of the point of contact and also the equation of the common tangent.

Sol. The centres of the given circle are A (2, 3) and B (–1, –1).

The radii of the circles are $r_1 = \sqrt{4 + 9 - 9} = 2$ and $r_2 = \sqrt{1 + 1 + 7} = 3$.

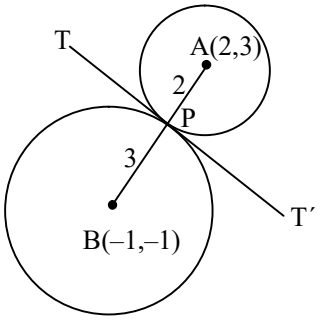
Furthermore $AB = \sqrt{(2+1)^2 + (3+1)^2} = 5$.

Since $r_1 + r_2 = 2 + 3 = 5 = AB$, the circles touch externally at a point P (α, β).

Further more AP : PB = 2 : 3 and so

$\alpha = \dfrac{2(-1)+3(2)}{2+3} = \dfrac{4}{5}$, $\beta = \dfrac{2(-1)+3(3)}{2+3} = \dfrac{7}{5}$

i.e. P is the point $\left(\dfrac{4}{5},\dfrac{7}{5}\right)$



Let TPT ' be the common tangent at P. Slope of PB is $\dfrac{(7/5)+1}{(4/5)+1} = \dfrac{4}{3}$

Hence the slope of TPT ' which is perpendicular to PB is – 3/4 therefore the equation of TPT ' is

$(y - 7/5) = -(3/4)(x - 4/5)$

i.e. $4(5y - 7) + 3(5x - 4) = 0$

or $3x + 4y - 8 = 0$

Ex.5 Find the equation of the circle which passes through the point (2, 0) and whose is the limit of the point of intersection of the lines $3x + 5y = 1$ and $(2 + c)x + 5c^2y = 1$ as c tends to 1.

Sol. Let A ≡ (2, 0)

given lines are $3x + 5y = 1$ ...(1)

$(2 + c)x + 5c^2y = 1$ ...(2)

multiplying equation (1) by c^2 and subtracting (2) from it, we get

$(3c^2 - c - 2)x = c^2 - 1$ or $x = \dfrac{c^2-1}{3c^2-c-2}$

Now $\lim\limits_{c\to 1} x = \lim\limits_{c\to 1} \dfrac{(c-1)(c+1)}{(c-1)(3c+2)}$

$= \lim\limits_{c\to 1} \dfrac{c+1}{3c+2} = \dfrac{2}{5}$

$\therefore$ x coordinate of centre $= \dfrac{2}{5}$

From (1), when $x = \dfrac{2}{5}$, $y = -\dfrac{1}{25}$

Hence centre of the circle is $\left(\dfrac{2}{5},-\dfrac{1}{25}\right)$

Also the circle passes through the point A (2, 0)

$\therefore$ radius of the circle

$= \sqrt{\left(2-\dfrac{2}{5}\right)^2\left(0+\dfrac{1}{25}\right)^2}$

Now equation of the required circle is

$\left(x-\dfrac{2}{5}\right)^2 + \left(y+\dfrac{1}{25}\right)^2 = \dfrac{64}{25} + \dfrac{1}{625}$

or $x^2 - \dfrac{4}{5}x + \dfrac{4}{25} + y^2 + \dfrac{2}{25}y + \dfrac{1}{625} = \dfrac{64}{25} + \dfrac{1}{625}$

or $25(x^2 + y^2) - 20x + 2y - 60 = 0$. **Ans.**

Ex.6 Show that each of the circles :

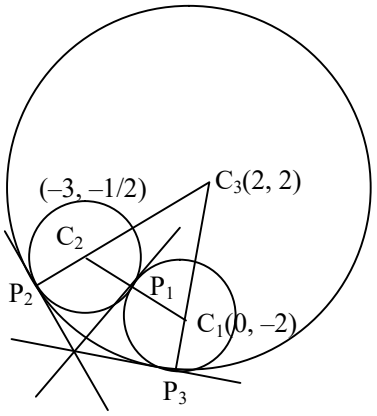
$S_1 \equiv x^2 + y^2 + 4y - 1 = 0$,

$S_2 \equiv x^2 + y^2 + 6x + y + 8 = 0$,

$S_3 \equiv x^2 + y^2 - 4x = 4y - 37 = 0$

touches the other two; determine the points of contact and the point of concurrence of the tangents at these points.

Sol. S_1 has centre C_1 (0, –2) and radius $r_1 = \sqrt{5}$;



S_2 has centre $C_2\left(-3,-\dfrac{1}{2}\right)$ and radius $r_2 = \dfrac{1}{2}\sqrt{5}$

S_3 has centre C_3 (2, 2) and radius $r_3 = 3\sqrt{5}$;

$C_1C_2 = \sqrt{9+\dfrac{9}{4}} = \dfrac{3}{2}\sqrt{5} = r_1 + r_2$

$\Rightarrow S_1$ and S_2 touch externally ...(1)

$C_2C_3 = \sqrt{25+\dfrac{25}{4}} = \dfrac{5}{2}\sqrt{5} = r_3 - r_2$

$\Rightarrow S_2$ and S_3 touch internally ...(2)

$C_3C_1 = \sqrt{4+16} = 2\sqrt{5} = r_3 - r_1$

$\Rightarrow S_3$ and S_1 touch internally ...(3)

From (1)

$\Rightarrow$ the point of contact P_1 divides C_1C_2 in the ratio $r_1 : r_2 = 2 : 1$

$\therefore$ P_1 is $(-2, -1)$.

From (2)

$\Rightarrow$ the point of contact P_2 divides C_2C_3 in the ratio $-r_2 : r_3 = -1 : 6$

$\therefore$ P_2 is $(-4, -1)$.

From (3)

$\Rightarrow$ the point of contact P_3 divides C_3C_1 in the ratio $r_3 : -r_2 = 3 : -1$

$\therefore$ P_3 is $(-1, -4)$.

the common tangent at $P_1 \equiv 2x - y + 3 = 0$

(i.e. $S_1 - S_2 = 0$) ...(4)

the common tangent at $P_2 \equiv 2x + y + 9 = 0$

(i.e. $S_2 - S_3 = 0$) ...(5)

the common tangent at $P_3 \equiv x + 2y + 9 = 0$

(i.e. $S_3 - S_1 = 0$) ...(6)

(4) and (5) intersect at $(-3, -3)$ which satisfies (6)

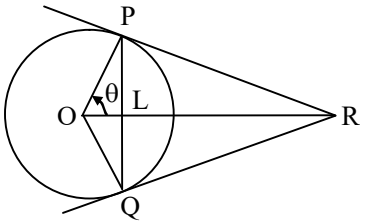
Hence $(-3, -3)$ is the point of concurrence of (4), (5) and (6). **Ans.**

Ex.7 Show that the area of the triangle formed by the pair of tangent from (x_1, y_1) to the circle $x^2 + y^2 = a^2$ and the chord joining their points of contact with the circle is a

$(x_1^2 + y_1^2 - a^2)^{3/2} (x_1^2 + y_1^2)^{-1}$ sq. units.

Sol. Let R (x_1, y_1) be the point from which are drawn the tangent RP and RQ to the circle $x^2 + y^2 = a^2$. Let OR meet PQ at L, where O (0, 0) is the centre of the circle.

Then $\Delta RPQ = 2\,\Delta RLP = 2\,[\Delta OPR - \Delta OLP]$



But $\Delta OPR = \frac{1}{2} OP.PR = \frac{1}{2}\, a.\sqrt{x_1^2 + y_1^2 - a^2}$

and $\Delta OLP = \frac{1}{2} (OP \cos\theta)\,(OP \sin\theta)$

$= \frac{1}{2} a^2 \cos\theta \sin\theta$ where $\angle PQR = \theta$

But from ΔOPR, $\sin\theta = \frac{PR}{OR}$

$= \frac{\sqrt{x_1^2 + y_1^2 - a^2}}{\sqrt{x_1^2 + y_1^2}}$

and $\cos\theta = \frac{OP}{OR} = \frac{a}{\sqrt{x_1^2 + y_1^2}}$

Hence $\Delta LOP = \frac{1}{2} a^2 \frac{\sqrt{x_1^2 + y_1^2 - a^2}}{\sqrt{x_1^2 + y_1^2}} \frac{a}{\sqrt{x_1^2 + y_1^2}}$

$= \frac{1}{2} a^3 \frac{\sqrt{x_1^2 + y_1^2 - a^2}}{x_1^2 + y_1^2}$

and ΔRPQ

$= 2\left[\frac{1}{2} a \sqrt{x_1^2 + y_1^2 - a^2} - \frac{1}{2} a^3 \frac{\sqrt{x_1^2 + y_1^2 - a^2}}{x_1^2 + y_1^2}\right]$

$= a\sqrt{x_1^2 + y_1^2 - a^2}\left[1 - \frac{a^2}{x_1^2 + y_1^2}\right]$

$= a\sqrt{x_1^2 + y_1^2 - a^2}\ \frac{(x_1^2 + y_1^2 - a^2)}{x_1^2 + y_1^2}$

$= \frac{a(x_1^2 + y_1^2 - a^2)^{3/2}}{x_1^2 + y_1^2}$ sq. units. **Ans.**

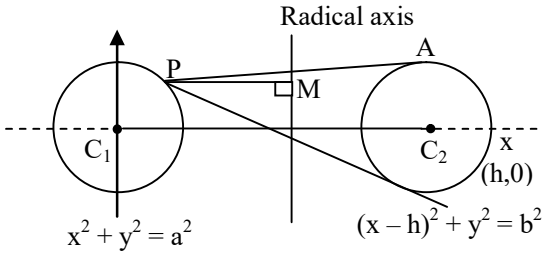
Ex.8 Prove that the square of the tangent that can be drawn from any point on one circle to another circle is equal to twice that product of the perpendicular distance of the point from the radical axis of the two circles, and the distance between their centres.

Sol. We have to prove that

$PA^2 = 2\,PM.\,C_1C_2$

$P \equiv (a\cos\theta, a\sin\theta)$

$PA = \sqrt{a^2 - b^2 + h^2 - 2ah\cos\theta}$



Radical axis :

$-2hx + h^2 - a^2 + b^2 = 0$

or $x = \frac{h^2 - a^2 + b^2}{2h}$

$$PM = \frac{h^2 - a^2 + b^2}{2h} - a\cos\theta$$

$$= \frac{h^2 - a^2 + b^2 - 2ah\cos\theta}{2h}$$

$$PM.\ C_1C_2 = \frac{h^2 - a^2 + b^2 - 2ah\cos\theta}{2h}$$

(where $C_1C_2 = h$)

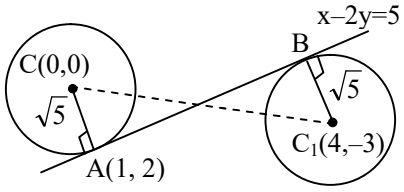
$$= \frac{PA^2}{2} \Rightarrow \therefore\ PA^2 = 2\ PM.\ C_1C_2$$ **Ans.**

Ex.9 Prove that the tangent to the circle $x^2 + y^2 = 5$ at the point (1, –2) also touches the circle $x^2 + y^2 – 8x + 6y + 20 = 0$ and find its point of contact.

Sol. $S \equiv x^2 + y^2 = 5$, centre C (0, 0), radius $\sqrt{5}$.

$S_1 \equiv x^2 + y^2 – 8x + 6y + 20 = 0$,

C_1 (4, – 3), $r_1 = \sqrt{5}$



S and S_1 have equal radii.

The equation of the tangent at A(1, – 2) to S = 0 is x – 2y = 5.

Since the distance of C_1 (4, –3) from this line $= \sqrt{5}$, which is equal to the radius of

$S_1 = 0$, this line touches $S_1 = 0$.

Let the point of contact be B.

Let line CC_1 cut this line at D. **Ans.**

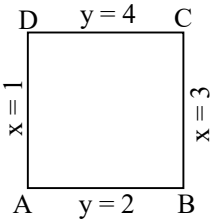
Ex.10 The sides of a square are x = 1, x = 3, y = 2 and y = 4. Find the equation of the circle drawn on the diagonals of the square as its diameter.

Sol. Let ABCD be a square and equation of its sides AD, BC, AB and DC be respectively x = 1, x = 3, y = 2 and y = 4,

Then A ≡ (1, 2), B ≡ (3, 2) ,

C ≡ (3, 4) and D ≡ (1, 4)

Now equation of the circle having AC as a diameter is



$(x – 1)(x – 3) + (y – 2)(y – 4) = 0$

or $x^2 + y^2 – 4x – 6y + 11 = 0$ ...(1)

(1) is also the equation of the circle having BD as its diameter. **Ans.**

Ex.11 The circle $x^2 + y^2 – 6x – 10y + k = 0$ does not touch or intersect the coordinate axes and the point (1, 4) is inside the circle. Find the ranges of the values of k.

Sol. $x^2 + y^2 – 6x – 10y + k = 0$.

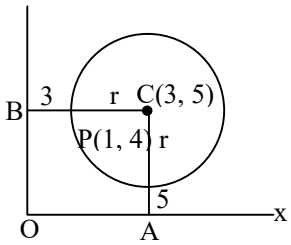
Centre C (3, 5), Point P (1, 4)

$r^2 = 9 + 25 – k = 34 – k$.

Point P lies inside the circle therefore

$CP^2 < r^2$

or $(3 – 1)^2 + (5 – 4)^2 < r^2$



or $4 + 1 < 34 < k$

or $– 29 + k < 0$ i.e. – ve

$\therefore\ k < 29$

If it touches the x - axis or intersect it, then radius is either equal to, or greater than y-coordinates of the centre i.e. 5.

Since it neither touches nor intersects therefore radius is less than 5.

Similarly, it does not touch or intersect y-axis therefore r < h the x coordinate of centre

$\therefore\ r < 3$ or $34 – k < 9$ or $25 < k$

i.e. $k > 25$

Hence the conditions are $k > 25$, $k < 29$ **Ans.**

Ex.12 Show that the equation of the image of the circle $x^2 + y^2 + 16x – 24y + 183 = 0$ by the line mirror $4x + 7y + 13 = 0$ is $x^2 + y^2 + 4x + 235 = 0$.

Sol. The given circle is (–8, 12), 5

The given line is $4x + 7y + 13 = 0$

The required circle while (h, k), 5 where (h, k) is the reflection of centre (–8, 12) in the line.

$4x + 7y + 13 = 0$

$$\therefore \frac{h-\alpha}{a} = \frac{k-\beta}{b} = -\frac{2\ell'}{a^2+b^2}$$

$$\text{or } \frac{h+8}{4} = \frac{k-12}{7} = -2$$

$\therefore h = -16, k = -2$

$\therefore$ Required circle is $(x + 16)^2 + (y + 2)^2 = 5^2$

or $x^2 + y^2 + 32x + 4y + 235 = 0$ **Ans.**

Ex.13 A circle touches both the x-axis and the line $4x – 3y + 4 = 0$. Its centre is in third quadrant and lies on the lines $x – y – 1 = 0$. Find the equation of the circles.

Sol. Let the equation of the required circle be

$x^2 + y^2 + 2gx + 2fy + c = 0$ ...(1)

Its centre is C (– g , – f) and radius is $\sqrt{g^2+f^2-c}$

Since circle (1) touches the x - axis

$\therefore g^2 – c = 0$ or $c = g^2$

$[\because \sqrt{g^2+f^2-c} = |f|]$ ...(2)

Again since circle (1) touches the line $4x – 3y + 4 = 0$ ...(3)

$$\therefore \frac{|-4g+3f+4|}{5} = \sqrt{g^2+f^2-c} = \sqrt{f^2} = |f|$$

[From (2)]

or $– 4g + 3f + 4 = \pm 5f$

$[\because |a| = |b| \Leftrightarrow a = \pm b]$

$$\therefore \begin{cases} 4g+2f=4 & \text{or } 2g+f=2 \quad ...(4) \\ -4g+8f=-4 & \text{or } g-2f=1 \quad ...(5) \end{cases}$$

Again since centre C (–g, – f) lies on the line $x – y – 1 = 0$

$\therefore – g + f = 1$ ...(6)

Solving (4) and (6), we get $g = 1/3, f = 4/3$

Thus C ≡ (– 1/3, – 4/3) which lies in the third quadrant.

Also from (2), $c = g^2 = 1/9$

Solving (5) and (6), we get $f = –2, g = – 3$

$\therefore$ c ≡ (3, 2) which lies in the first quadrant.

Thus for the required circle $g = 1/3$, $f = 4/3$, $c = 1/9$

$\therefore$ Equation of the required circle will be

$$x^2 + y^2 + \frac{2}{3}x + \frac{8}{3}y + \frac{1}{9} = 0$$

or $9(x^2 + y^2) + 6x + 24y + 1 = 0$ **Ans.**

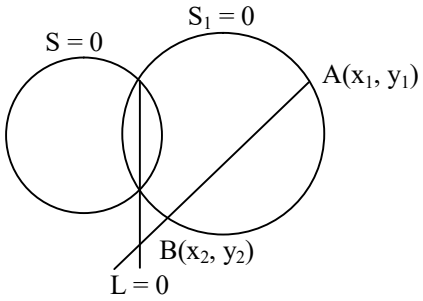
Ex.14 A fixed circle is cut by a family of circles all of which pass through two given points $A(x_1, y_1)$ and $B(x_2, y_2)$. Prove that the chord of intersection of the fixed circle with any circle of the family passes through a fixed point.

Sol. Let S = 0 be the equation of the fixed circle.

Let $S_1 = 0$ be the equation of any circle through A and B which intersect S = 0 in real points.

$L \equiv S – S_1 = 0$ is the equation of the chord of intersection of S = 0 and $S_1 = 0$.

Let $L_1 = 0$ be the equation of the line AB.



Then $S_1 + \lambda L_1 = 0$ is the equation of the family of circles all of which pass through A and B.

The circles of this family which cuts the fixed circle S = 0 in real points, give us a system of the chord of intersection.

The equation of the chords of this system are of the type :

$S – (S_1 + \lambda L_1) = 0$ or $(S – S_1) – \lambda L_1 = 0$

or $L – \lambda L_1 = 0$ ...(1)

(1) implies each chord of intersection passes through the fixed point - which is the point of intersection of lines L = 0 and $L_1 = 0$.

Hence proved.

Ex.15 Show that the equation $x^2 + y^2 – 2x – 2ay – 8 = 0$ represents, for different values of ' a ' a system of circles passing through two fixed point A, B on the x - axis, and find the equation of that circle of the system the tangents to which at A, B meet on the line $x + 2y + 5 = 0$.

Sol. The given equation can be written in the form $(x^2 + y^2 – 2x – 8) + a(– 2y) = 0$, is the equation of a system of circles passing through the points of intersection of the fixed circle $x^2 + y^2 – 2x – 8 = 0$ and the fixed line $y = 0$ (i.e. x-axis).

These points of intersection are therefore fixed, they are given by

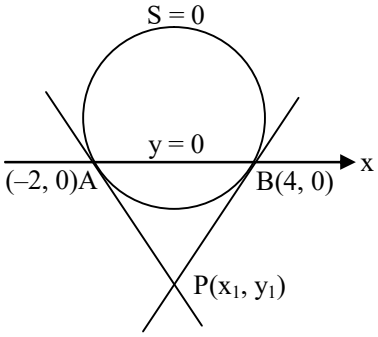
$x^2 – 2x – 8 = 0$ or $(x + 2)(x – 4) = 0$

and therefore they are A $(–2, 0)$ and B $(4, 0)$

Let the circle of the system to which the tangents at A and B meet on the line $x + 2y + 5 = 0$, be given by $a = k$.

Then it is :

$S \equiv x^2 + y^2 – x – 2ky – 8 = 0$ ...(1)



Let the tangent at A and B to (1) intersect at P (x_1, y_1). Then AB is the chord of contact given by the tangents drawn from P (x_1, y_1) to the circle $S = 0$.

The equation of AB should be

$xx_1 + yy_1 – (x + x_1) – k(y + y_1) – 8 = 0$ ...(2)

but the equation AB is $y = 0$ ...(3)

(2) and (3) being identical $x_1 – 1 = 0$ and $x_1 + ky_1 + 8 = 0$

$\therefore x_1 = 1$ and $y_1 = – 9/k$

$\therefore P(x_1, y_1) = P(1, – 9/k)$

But we are given that P (x_1, y_1) lies on the line $x + 2y + 5 = 0$

$\therefore 1 – 18/k + 5 = 0$

$\therefore k = 3$

Hence the required circle given by (1) is $x^2 + y^2 – 2x – 6y – 8 = 0$. **Ans.**

Ex.16 Show that the line $3x – 4y – c = 0$ will meet the circle having centre at (2, 4) and the radius 5 in real and distinct points if $– 35 < c < 15$.

Sol. Given line is $3x – 4y – c = 0$ ...(1)

Centre of given circle is (2, 4) and its radius is 5, therefore its equation will be

$(x – 2)^2 + (y – 4)^2 = 5^2$

or $x^2 + y^2 – 4x – 8y – 5 = 0$ ...(2)

From (1) $y = \frac{1}{4}(3x – c)$. Putting the value of y in (2), we get

$x^2 + \frac{1}{16}(3x – c)^2 – 4x – 8.\frac{1}{4}(3x – c) – 5 = 0$

or $16x^2 + 9x^2 – 6cx + c^2 – 64x$

$– 96x + 32c – 80 = 0$

or $25x^2 – 2(80 + 3c)x + c^2 + 32c – 80 = 0$ ...(3)

Line (1) will meet the circle (2) in real and distinct points if discriminant of equation (3) > 0

i.e. if $4(80 + 3c)^2 – 100(c^2 + 32c – 80) > 0$

or $(80 + 3c)^2 – 25(c^2 + 32c – 80) > 0$

or $6400 + 9c^2 + 480c – 25c^2 – 800c + 2000 > 0$

or $– 16c^2 – 320c + 8400 > 0$

or $16c^2 + 320c – 8400 < 0$

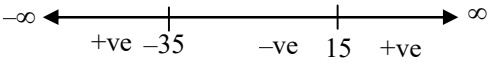
or $c^2 + 20c – 525 < 0$

Sign scheme for $c^2 + 20c – 525$:

When $c^2 + 20c – 525 = 0$,

$$c = \frac{–20 \pm \sqrt{400 + 2100}}{2} = \frac{–20 \pm 50}{2} = – 35, 15$$

Therefore, sign scheme for $c^2 + 20c – 525$ is as follows



$\therefore c^2 + 20c – 525 < 0 \Leftrightarrow – 35 < c < 15$ **Ans.**

Ex.17 Find the equation of the circle of minimum radius which contains the three circles

$x^2 + y^2 – 4y – 5 = 0$,

$x^2 + y^2 + 12x + 4y + 31 = 0$ and

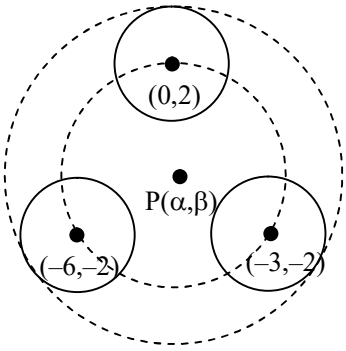
$x^2 + y^2 + 6x + 12y + 36 = 0$

Sol. For the first circle,

centre = (0, 2) and radius = $\sqrt{(–2)^2 + 5} = 3$.

For the second circle,

centre (–6, –2) and radius = $\sqrt{6^2 + 2^2 – 31} = 3$



For the third circle,

centre = (–3, –6) and radius = $\sqrt{3^2+6^2-36}=3$.

Let P (α, β) be the centre of the circle passing through the centres of the three given circles. Then

$\alpha^2+(\beta-2)^2=(\alpha+3)^2+(\beta+6)^2$

$=(\alpha+6)^2+(\beta+2)^2$, {$= r^2$}

or $4-4\beta=6\alpha+9+12\beta+36$

$=12\alpha+36+4\beta+4$

$\therefore\ 6\alpha+16\beta+41=0$ ...(1)

and $6\alpha-8\beta-5=0$ ...(2)

(1) – (2) $\Rightarrow\ 24\beta+46=0$

$\therefore\ \beta=-23/12$

$\therefore$ (2) $\Rightarrow 6\alpha-8.(-23/12)-5=0$

or $6\alpha+46/2-5=0$; $\therefore\ \alpha=-31/18$

$\therefore$ the centre of the circle passing through centres of given circle is

(– 31/18, – 23/12) and

radius $r=\sqrt{\alpha^2+(\beta-2)^2}$

$$=\sqrt{\left(-\frac{31}{18}\right)^2+\left(-\frac{23}{12}-2\right)^2}$$

$$=\sqrt{\frac{31^2}{18^2}+\frac{47^2}{12^2}}=\sqrt{\frac{4\times31^2+9\times47^2}{36^2}}$$

$$=\frac{1}{36}\sqrt{23725}=\frac{5}{36}\sqrt{949}.$$

$\therefore$ the radius of the required circle

$=3+\frac{5}{36}\sqrt{949}$ and centre = (– 31/18, – 23/12)

$\therefore$ the equation of the required circle is

$$\left(x+\frac{31}{18}\right)^2+\left(y+\frac{23}{12}\right)^2=\left(3+\frac{5}{36}\sqrt{949}\right)^2$$

Ans.

Ex.18 If $4\ell^2-5m^2+6\ell+1=0$ then show that the line $\ell x+my+1=0$ touches a fixed circle. Find the radius and centre of the circle.

Sol. Let the circle be $(x-\alpha)^2+(y-\beta)^2=a^2$.

The line $\ell x+my+1=0$ touches the circle if

$$a=\left|\frac{\ell\alpha+m\beta+1}{\sqrt{\ell^2+m^2}}\right|$$

or $a^2(\ell^2+m^2)=(\ell\alpha+m\beta+1)^2$

or $(a^2-\alpha^2)k^2+(a^2-\beta^2)m^2$

$-2\alpha\beta\ell m-2\ell\alpha-2m\beta-1=0$ ...(1)

But $4\ell^2-5m^2+6\ell+1=0$ ...(2)

It is possible to find α, β, a if (1) and (2) are identical.

The condition is

$$\frac{a^2-\alpha^2}{4}=\frac{a^2-\beta^2}{-5}=\frac{-2\alpha\beta}{0}=\frac{-2\alpha}{6}=\frac{-2\beta}{0}=\frac{-1}{1}$$ ...(3)

$\therefore\ \beta=0,\ \alpha=3$

and $\frac{a^2-\alpha^2}{4}=-1$ which implies

$a^2-3^2=-4$, i.e., $a=\sqrt{5}$.

Also $\alpha=3,\ \beta=0,\ a=\sqrt{5}$ satisfies (3)

$\therefore$ the line touches the fixed circle,

$(x-3)^2+(y-0)^2=(\sqrt{5})^2$

or $x^2+y^2-6x+4=0$

whose centre = (α, β) = (3, 0) and radius

$=a=\sqrt{5}$. **Ans.**

Ex.19 Find the equation of the circles passing through (–4, 3) and touching the lines x + y = 2 and x – y = 2.

Sol. The centre of the circle C is equidistant from the two tangents ; x + y = 2 and x – y = 2.

$\therefore$ It lies on the bisector of the angle between them which is y = 0.

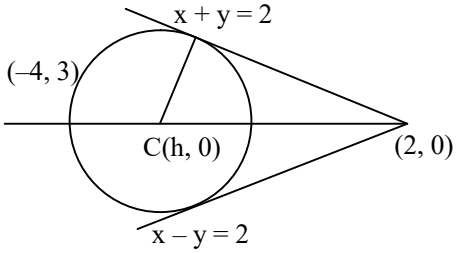
[The other bisector x = 2 is to be rejected

$\because$ (–4, 3) a point in the 2nd quadrant lies on the circle]

$\therefore$ Let C ≡ (h, 0)

The circle passes through (–4, 3)

$\therefore (h+4)^2 + 3^2 = \left[\left|\frac{h-2}{\sqrt{2}}\right|\right]^2$



$\Rightarrow \quad h^2 + 20h + 46 = 0$

$\therefore \quad h = -10 \pm 3\sqrt{6}$

$\therefore$ There are 2 circles.

Circle I :

$C_1 \equiv (-10 + 3\sqrt{6}\ , 0)$ and

$r_1 = \left|\frac{-10 + 3\sqrt{6} - 2}{\sqrt{2}}\right|$

$\therefore$ Its equation is $(x + 10 - 3\sqrt{6})^2 + y^2$

$= (6\sqrt{2} - 3\sqrt{3})^2$

Circle II :

$C_2 \equiv (-10 - 3\sqrt{6}\ , 0)$ and $r_2 = 6\sqrt{2} + 3\sqrt{3}$

$\therefore$ The equation of the circle is

$(x + 10 + 3\sqrt{6})^2 + y^2 = (6\sqrt{2} + 3\sqrt{3})^2$

Ans.

Ex.20 Find the equations to the common tangents to the circles $x^2 + y^2 = 25$ and $(x - 12)^2 + y^2 = 9$.

Sol. For $x^2 + y^2 = 25$ $C_1 \equiv (0, 0)$; $r_1 = 5$

$(x - 2)^2 + y^2 = 9$ $C_2 \equiv (12, 0)$; $r_2 = 3$

$d(C_1 C_2) = 12$

$d(C_1 C_2) > r_1 + r_2$.

$\therefore$ The two circles lie external to each other.

$\therefore$ There shall be four common tangents to the circles, two direct and two transverse.

Let $C_1 C_2$ be divided internally and externally in the ratio 5 : 3.

Internally at the point (15/2, 0) and externally at the point (30, 0).

The straight line through (15/2, 0) with a slope m is $y = m(x - 15/2)$ i.e. $2mx - 2y - 15m = 0$

This straight line will be a tangent to the circle $x^2 + y^2 = 25$

if $\left|\frac{15m}{\sqrt{4m^2 + 4}}\right| = 5$

i.e. if $m = \pm \frac{2}{\sqrt{5}}$

$\therefore$ The transverse common tangents are given by $y = \pm \frac{2}{\sqrt{5}}(x - 15/2)$

Now the straight line through (30, 0) with a slope m is given by

$y = m'(x - 30)$

i.e. $m'x - y - 30\,m' = 0$

This straight line will be a tangent to the circle $x^2 + y^2 = 25$

if $\left|\frac{15m'}{\sqrt{m'^2 + 1}}\right| = 5$; i.e. if $m' = \pm \frac{1}{\sqrt{35}}$

$\therefore$ The equation of the two direct common tangents are

$y = \pm \frac{1}{\sqrt{35}}(x - 30)$ **Ans.**

Ex.21 Let $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ be a given circle. Find the locus of the foot of the perpendicular drawn from the origin upon any chord of S which subtends a right-angle at the origin.

Sol. Let $\ell x + my = 1$ be any chord of $S = 0$ which subtends a right - angle at O.

Let A, B be the point of intersection of the chord and circle.

The join equation of the line OA and OB can be obtained by homogenizing the equation $S = 0$ with the help of $\ell x + my = 1$.

Therefore their joining equation is :

$x^2 + y^2 + (2gx + 2fy)(\ell x + my) + c(\ell x + my)^2 = 0$ ...(1)

Since the angle between OA and OB is 90º.

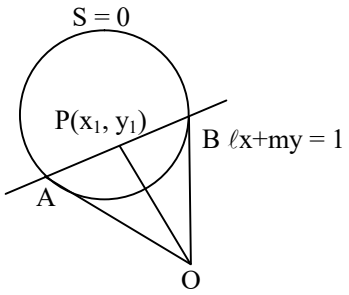
$\therefore$ The sum of coefficient of $x^2 + y^2$ in (1) = 0

This $\Rightarrow 2 + 2g\ell + 2fm + c(\ell^2 + m^2) = 0$ ...(2)

Let P (x_1, y_1) be the foot of perpendicular from O on AB.

Then $\ell x_1 + my_1 = 1$ and $mx_1 = \ell y_1$.

$\therefore \ell = \dfrac{x_1}{x_1^2 + y_1^2}$ and $m = \dfrac{y_1}{x_1^2 + y_1^2}$



Substituting in (2) we get,

$$2 + 2\frac{gx_1 + fy_1}{x_1^2 + y_1^2} + c\frac{x_1^2 + y_1^2}{(x_1^2 + y_1^2)^2} = 0$$

or $2(x_1^2 + y_1^2) + 2(gx_1 + fy_1) + c = 0$

Hence the locus of $P(x_1, y_1)$ is

$2(x^2 + y^2) + 2(gx + fy) + c = 0$. **Ans.**

Ex.22 If the origin is one of the limiting point of a system coaxial circles of which $x^2 + y^2 + 2gx + 2fy + c = 0$ is a member, find the coordinates of the other limiting point.

Sol. One limiting point is the origin, namely the point circle $(x - 0)^2 + (y - 0)^2 = 0$

i.e. $x^2 + y^2 = 0$

This is one member of the coaxial system and another member is

$x^2 + y^2 + 2gx + 2fy + c = 0$

Hence the equation of the radical axis of the system is

$(x^2 + y^2 + 2gx + 2fy + c) - (x^2 + y^2) = 0$

i.e. $2gx + 2fy + c = 0$

The equation of the coaxial system is thus

$x^2 + y^2 + \lambda (2gx + 2fy + c) = 0$

Its centre is $(-g\lambda, -f\lambda)$ and its radius $\sqrt{g^2\lambda^2 + f^2\lambda^2 - c\lambda}$. For a limiting point, the radius is 0.

Hence $g^2 \lambda^2 + f^2\lambda^2 - c\lambda = 0$, giving

$\lambda [\lambda(g^2 + f^2) - c] = 0$, so that $\lambda = 0$,

$$\lambda = \frac{c}{g^2 + f^2}$$

One limiting point is (0, 0) and the other is

$$\left(-\frac{gc}{g^2 + f^2}, -\frac{fc}{g^2 + f^2}\right)$$ **Ans.**